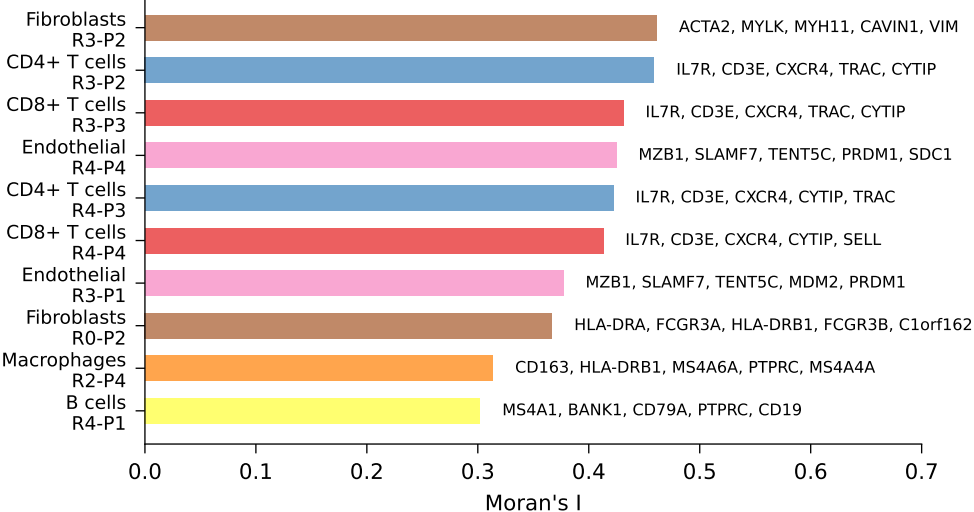


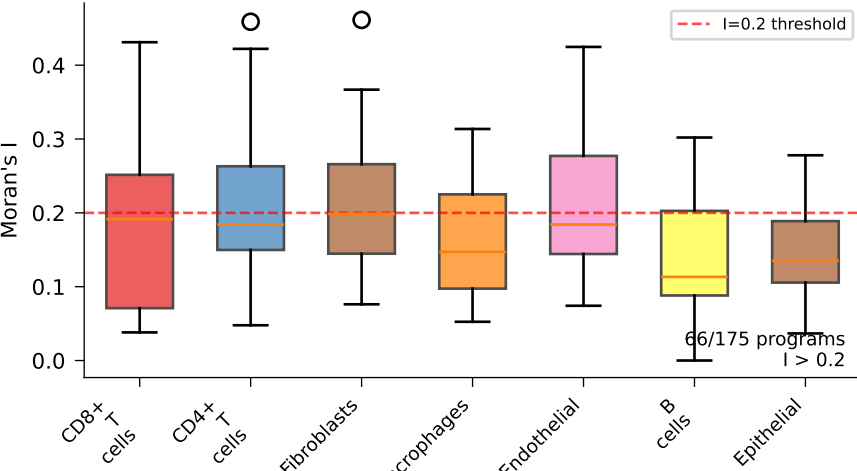
A. Protein-Anchored Program Discovery

Panel A: NMF Schematic
(Create in vector editor)

B. Top Spatially Coherent Programs



C. Spatial Coherence by Cell Type



D. Xenium Single-Cell Resolution
(5 regions, 7 cell types, 175 programs)

	Mean I	Max I	N Programs	Mean Var%
Region 0	0.146	0.367	35.0	14.396
Region 1	0.127	0.315	35.0	14.413
Region 2	0.193	0.377	35.0	12.96
Region 3	0.226	0.461	35.0	14.436
Region 4	0.206	0.425	35.0	13.674

Same Module 4 workflow at single-cell resolution

E. Program Spatial Relationships

